Plexus Discovery

Progress note — written to be read, not decoded

31 July 2026

Plain English throughout. Organised by phase, because phases are where we stop and you decide.

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1. What we are doing — two things at once

Track A — build the method. An automated loop that searches for biological *mechanisms* rather than parameter values. It proposes a change to the model's structure ("remove the pushing force", "add oriented cell division"), writes down what it expects *before* running, runs the simulation, measures, and records whether it was right. The discipline that makes it worth building: a change of numbers can never masquerade as a new idea.

Track B — reproduce Okuda's result, as the proof that Track A works. Okuda et al. (2018) grew a hollow ball of cells that spontaneously forms tubes, undulations and branches, driven by a chemical pattern on its own surface. We are rebuilding that here. If the loop finds the mechanism largely by itself, the method is demonstrated. *The figure is not a side quest — it is the evidence.*

The two tracks feed each other: the reproduction is the honest test of the method, and the method is what makes the reproduction more than hand-tuning.

2. Where we actually are

Track A is further along than Track B, and every number taken before 31 July was measured with at least one instrument since proved wrong. So the register of results is empty on purpose, and the work right now is rebuilding the instruments rather than running experiments.

3. What we believe about the biology

Nothing *measured*. But we now write down what we assume.

The register of *results* is still empty, and should stay empty until a batch has been taken with instruments that work. Every number we have was measured with at least one broken ruler in the path.

What changed on 31 July is a different thing: the register of **assumptions** is no longer empty. Eleven basics about cell tissue are written down in `discovery/PREMISES.md` — not a literature review, but the minimum a person needs in order to look at one of our runs and say "that cannot be right". Each is phrased so that a computer can check it against a run, and each one now does (Phase 1).

The distinction matters. A result is something we found out. A premise is something we are *assuming*, stated plainly enough that it can be caught being wrong. Writing the second kind down is not a claim about biology — it is a way of making our own mistakes visible.

4. Two questions instead of one number

Success is not a threshold a single run clears. It is two questions, and they are asked separately because they fail separately.

1. What can we reproduce?	Each of the paper's morphologies is a row, scored qualitatively: *not attempted / attempted and failed / partial / reproduced.* **Today 0 of 4.** Budding is tracked too — our substrate produces it and no criterion ever mentioned it.
2. What do we know?	Already measured, and previously subordinate to the tube gate: map coverage **27%** (3 of 11 cells can state a verdict) — and all of it is routing; solo operator effects are **0 of 8**. Nine claims resolved, ten open, surprise rate **0.375**.

5. What is genuinely working

Four things have earned their keep, and they are all about catching ourselves rather than about tissue.

Predictions are written before runs, and wrong ones stay wrong. The findings register now holds **six standing claims and six retracted**, each retraction carrying the reason it was believed in the first place. Nothing is quietly edited away.

Instruments are certified against a known answer before they are used. This already cost one of my own: a wavelength measure failed its own calibration — the ratio it depended on came out 1.37, 1.79, 1.49 instead of a constant — and it was demoted to a labelled diagnostic instead of being shipped. A metric that fails certification does not get to pass.

The premise gate has caught something on its own. A smoke run was refused before any result was read, because the chemistry had been extinguished by growth dilution. That is the first time anything in this project stopped a bad run without a human looking at a picture.

The loop refuses to guess. Asked to score 32 results against predictions written in a unit it did not recognise, it declined 32 times rather than inventing 32 confirmations.

The standing caveat. Every defect found so far — and there have been more than twenty —

was found by Cedric looking at a picture. The gate above is the first thing built to change that, and it has one catch to its name. That number is the honest measure of Track A.

6. The team of agents

Nine agents read evidence and form a judgement; five checks are deterministic code that simply refuses. Eight of the nine now run every round — the Grounder was wired this session, and until then nothing in the loop had ever asked the paper anything, which is why a 27-run batch started at 150 cells against a paper that says 200. Evolution was wired in the same pass; only the Referee's tournament remains idle.

Figure 1: *
The team. Every agent now runs: the **Grounder** and **Evolution** were written and never called until 1 August, and the Grounder's absence is why a batch started at 150 cells against a paper that says 2,000. Only the **Referee** is still idle.

Agents — they read evidence and form a judgement				
Proposer	chooses which mechanism edits to test next; also checks the batch against what has already been tried	1	not yet	runs
Peer-review	reads the proposed batch before it costs anything	1	not yet	runs
Analysts (×3)	read one run each, independently, and say what happened	15	**21 s** each	runs
Eye-check	watches the movie; vetoes a number the picture contradicts	5	not yet	runs
Judge	second opinion on the top pair	1	not yet	runs
Interpreter	writes the causal description into the record	2	not yet	runs
Meta-review	distils the round into patterns worth carrying forward	1	not yet	runs
Grounder	**reads Okuda's paper; sets each batch's starting conditions from it, and refuses to advise if its own quotes do not check out**	1	not yet	runs
Evolution	refines the round's winner, rather than the loop only ever picking one	1	not yet	runs
Checks — deterministic code that refuses. No judgement, no cost.				
Supervisor	holds the objective and the budget; decides when to stop	—	—	runs
Critic	type-checks the batch; rejects a causal claim with no ablation; refuses a composition already tried	—	—	runs
Biologist	refuses a recipe that could not be a tissue	—	—	runs
Metrologist	certifies instruments; files defects and retractions; refuses a duplicate operator request	—	—	runs
Referee	rank a batch by tournament, not by sorting on one number	—	—	self-test only
Cedric	looks at the pictures; found every defect so far	—	—	outside the loop
one round	20 of the 26 calls are the per-run readers, and they are independent — run together, the block costs its slowest call	26	**64 s** per run	

Only the analysts have been measured, and the gap is large. Timing was wired on 1 August — until then nine call sites bypassed the stopwatch, and a round once reported `calls: 0` for a round that made about twenty-five. Six real analyst calls: **21 seconds** each against a four-minute allowance, and **7.2 turns** against a cap of 60. So neither the timeout nor the turn cap has ever constrained anything, and the "two hours a round" arithmetic I put to Cedric earlier was a ceiling nothing approaches. The other roles say *not yet* rather than carrying their allowance, because an allowance printed in a measurement column is how a ceiling becomes a fact.

7. What stands between us and the Okuda figure

Most of the gap is mundane and it is about starting conditions. We were running 150 cells where Okuda starts at 200, 2000 and 4000, and because the mesh buffer is sized as a multiple of the starting count, beginning at 150 capped every run at 1778 cells — so his largest case was not reachable at all. Two of his settings also sat outside the box our search was allowed to use: how sharply a cell switches on its growth (he uses 10, we allowed up to 8) and how much further the inhibitor travels than the activator (he uses 10, we allowed 2), and the second is why we could not produce his handful of well-separated spots. A third number, his pattern length-scale, is not even the same quantity as ours, so the decision taken with Cedric is to calibrate against what he *reports seeing* — the spot count on a ball of a given size — rather than copy a value across two differently-scaled models. All of that is four numbers and a buffer, and it is closed in Phase 2. The one genuine gap is that the coupling only runs one way: chemistry deforms the tissue, but the tissue's shape never feeds back into the chemistry, because our diffusion averages neighbours equally instead of weighting by shared wall area and cell volume. A plain tube probably does not need that arrow — a tube needs a spot that stays put, and a decal does that perfectly well. Branching almost certainly does need it, because in Okuda's account a tube forks when the tip stretches, dilutes its chemical unevenly, and the domain splits. So without it we would not get branching, and — worse — we would not know whether that was a real result or our missing arrow.

8. The phases

Eight stages. **I stop at the end of each one and wait for you.** That is the partition.

Phase 0 — Pre-flight: fix the instruments done (28 of 28)

Goal. Make every measurement trustworthy. Nothing runs unattended until this lands. Not one of the twenty-eight items is science; each one makes a question *askable.*

The last two, closed. *Comparing a batch at a common frame* — the per-run evidence horizon was right and did not make a BATCH comparable. If one run tears at frame 300 and another holds to 700, comparing their "final" states puts a torn mesh beside a healthy one and reports the difference as biology: both operands valid, the comparison wrong, which is the shape of every error that has cost this campaign a batch. A batch is now read at one frame, the earliest horizon in it, and the tool reports what that costs — how much of each run is discarded, and a warning when one run's early death is being imposed on everyone. *The explicit vocabulary* — `elongation`, `elongation_peak`, `elongation_at_end`, `elongation_unforced`. Writers emit both names, readers resolve either, the archive is never rewritten: cutting 92 references over in one step leaves a window where a key is written by half the code and read by the other half, and since an absent metric reads as "not measured", the bug would surface as a round that quietly learns nothing.

The repairs worth knowing about

what	what was found
survival test	It relaxed a fresh sphere and returned 1.014 for every run. Now real: **1.415 → 1.325, so 94% of the elongation survived the drivers going off** — the first honest survival number the campaign has produced.
the camera	Growth was drawn as *shrinkage*: apparent size fell to 0.52× on a run that grew by half. A second viewpoint justified itself immediately — in one run **the tube is invisible from the side and obvious from the top**.
the tube bar	**No capsule of any aspect ratio can reach the old bar of 2.0** (ceiling 1.90). The criterion was not asking for a tube; it was asking for the spike artefact.
damage	The blended damage score was counting *cell division*: sliver correlates with tip count at +0.94, while genuinely broken cells stayed at **0 for 300 frames**.
the 1778 ceiling	Arithmetic, not physics: (3552+4)/2 = 1778, and all 32 runs of the overnight study stopped there.
the missing arrow	The emitter never passed the implementation through — the ablation would have reported "no effect" *without ever running the new code.*
the curve plot	Drawn every run since the beginning, referenced **nowhere in the codebase**.
timing	The round reported `calls: 0` for a round that made about 25.

Phase 1 — Teach the loop what a tissue is: the premise gate — done

Why this became its own phase. Ten defects were found in one day, every one by Cedric looking at a picture, and all of the same kind: an operator or an instrument not doing what its name says. The loop checked that runs finished, that the mesh was valid, that the numbers were finite. It never once checked that the thing it had simulated was a tissue.

Eleven basics, each one something a computer can run and fail, in three tiers by cost: read the recipe (instant, refuses a hopeless run before a GPU is touched); read the run (free, goes loudly into the record); and run a small experiment of its own — switch everything off and let the ball sit for forty frames; it must not change size.

A broken premise must be declared, not silenced. Switching off growth to ask what a protrusion is made of genuinely breaks premise 1, and that is science. So a recipe may waive a premise *in writing, with a reason*; an undeclared violation is a hard stop. That is the only thing separating "we tested an idea" from "the settings were wrong".

Every check was proved to fail before being trusted. The one to look at: run against the code as it stood that morning, a resting ball collapses **5.00 → 0.69** in forty frames. That single check, which costs about a minute, would have caught the worst defect of the campaign on day one.

Phase 2 — Make Okuda's settings reachable — done (6 of 6)

The box used to describe the work without listing it, so the count had no visible denominator. All six, with what each one turned out to be:

. item	state	what it turned out to be
0. The ball could not get big enough	**done**	*Not on the list.* A spring held the shell at the size it was first built at while the cells inside grew sixteen-fold, so the sheet folded through itself. The spring's target now follows what the cells are asking for: the same run grows smoothly to 5 240 cells and stays clean.
1. Widen the out-of-range settings	**done**	Done, and the new bounds are *derived* rather than chosen: growth sharpness to 49.5 (Okuda needs 10), inhibitor spread to 12.5 (he needs 10), reaction speed unbounded.
2. Replace the mislabelled length-scale	**done**	The quantity we called a pattern size is a solver rate. Resolved in Phase 4 by measuring the pattern instead of setting it.
3. The shape-to-chemistry feedback	**done**	One contract, four implementations — curvature, tension, apical area, pressure — because *which* feature the chemistry listens to is the hypothesis, not a dial. Force and size were refused with reasons.
4. A fixture asserting his published constants	**done**	His Table 1 is now written down in the Grounder, quoted, with the sentence each value comes from — the Hill coefficient, the two diffusivities, the cell cycle, and the two regime *inequalities* that actually define his setting. Asserting a run against it is Phase 3, because it needs the Grounder wired.
5. His cell counts, with the buffer sized for the *destination*	**done**	**The item that voided the weekend batch**, and it is measured rather than argued — see below.

Item 5, tested rather than argued. A closed cell sheet is trivalent, so Euler fixes the vertex count at twice the cell count: a reservoir of 3552 vertices caps the tissue at $(3552+4)/2 = 1778$ cells exactly, whatever the biology wants. The reservoir is now computed from the count a run is *aiming at.* The same recipe, run twice, with nothing changed but the seed count and the reservoir:

	seeds	vertex buffer	cells at end	verdict
before	150	3 552	**1778**	saturated — not evidence
after	200	10 404	**4151**	clean — valid evidence

The old configuration stops dead on the rail; the new one grows past Okuda's largest case. Elongation is 1.061 against 1.073, so the physics is unchanged — the run simply gets to happen. Sizing the buffer from the seed instead of the destination is what produced 1778 twice in one week.

One error the check caught on me. My first version sized the reservoir from the seed count, which gave the 200-cell case a ceiling of 264. Okuda is explicit that his largest tissues were "picked up on the growth process" — a destination, not a start — so the two are now recorded as separate numbers.

Certification caught two of my own errors on item 3 — a curvature normalisation that only looked right because the test held cell count fixed, and a test of mine that demanded the wrong answer.

The phase closes on a green self-test, which it did not have an hour ago. The composition space checks itself in 61 ways, and eight were failing — silently, since yesterday. Seven were the growth-ceiling fix and the chemistry-clock fix diverging from the archived recipes: the space is right and the archive is wrong, so both are now *declared* deliberate rather than back-fitted into the archive, because rewriting a reference to make a test pass is how it stops being one. The eighth was real — no reference recipe used the new shape-to-chemistry operator, so the recipe named for Okuda's route was missing the half that makes it his. It carries it now. And the chemistry clock, previously exempt and unchecked, gained a test of its own: both terms must scale by exactly 1/dt *and* their ratio must be unchanged, since that ratio is what sets the Turing wavelength. **61 of 61.**

Phase 3 — Switch on the agents that never run done (3 of 3)

Ablation is now compulsory — *done*, and it is the part that matters. A claim that one thing *causes* another needs two experiments and we were enforcing neither: adding a mechanism and seeing the shape appear shows it is *sufficient*; taking it away and seeing the shape die shows it is *necessary*. Either alone is an opinion.

Every claim must now declare its kind, and the checker throws out a batch that asks for more than it tests — before any simulation starts. Sufficient earns "is sufficient for"; necessary earns "is required for"; both earn "causes"; neither earns "is associated with". **The word can no longer outrun the experiment.**

Still to do — and one of the three is now the priority.

(a) Wire the Grounder — *done*. It is the only agent that reads the paper and it had no call site anywhere; every entry point was exercised only by its own smoke test. Batch construction now asks it, and it answers with Okuda's own numbers: 2,000 cells, carrying the sentence they came from. It arrives as a *parameter*, so composition identity is unchanged — setting the starting size is a retune, not a new mechanism, and the search cannot mistake one for the other.

Two rules keep it from becoming a straitjacket. The Grounder *advises*: the faithful share of a batch inherits his numbers, the exploratory share is free to leave them — a search that may only ever stand where the paper stands cannot discover that the paper is not the only place to stand. What is *not* optional is the reservoir, and that lives in the translator where it applies to both shares alike.

And it was wrong the first time, in the most dangerous way available. My first version started tubulation at 200 cells, quoting the paper. The sentence was real; the mapping was not. "About 200 cells" belongs to two *control* experiments Okuda runs first — one on arrested tissue that cannot grow at all, one with growth decoupled from the morphogen. His three morphology figures are a single coupling experiment at **2,000** cells, and only χ and γ separate them. Cedric caught it. A quotation is not a citation: the quote made a tenfold error look verified, which is worse than having offered no quote at all. There is now no case to infer — a slot either declares which of Okuda's experiments it is, or it gets the coupling one.

Still open here: asserting a run against his published constants. The values are written down; nothing yet checks a run against them.

(b) Evolution and the Referee — *done*. `rank_btl` had been written and self-tested since the loop was built and had never ranked a real batch; the round sorted on whichever scalar the instrument gate happened to admit. A tournament decides now. Verified on a synthetic batch: a vetoed run holding the highest score drops from first to last. Evolution runs too, refining the winner rather than the loop only ever *picking* from what the enumerator offered.

(c) The per-run calls run together — *done, and measured*. Three analysts read one run concurrently, and runs are read concurrently across a batch. They are independent by design: three readings exist because one is not reproducible, and no analyst's prompt depends on another's answer. The eye-check still runs *after*, because it is handed their consensus and is not independent — parallelising a dependency is not an optimisation, it is a different experiment.

Phase 4 — Measurements that can tell his regimes apart done

The classifier — *done.* Sphere, undulation, tube, branched, plus **invalid** (the surface passes through itself, so no shape reading means anything) and **unclear**, returned deliberately near a boundary. A classifier that always answers can never be wrong, and one that can never be wrong teaches nothing. "Invalid" beats every morphology — this campaign has already once reported a crumpled shell as a result. It records the *path* a run takes, not just where it ends.
The calibration, and the surprise. The plan was to tune until we saw Okuda's handful of spots on a 2000-cell ball, then freeze. Counting spots over time — for the first time — showed something else:

step	0	50	100	**200**	400	3000
spots	104	62	19	**7**	1	1
coverage	0.06	0.14	0.37	0.49	0.52	0.53

The pattern never settles. It keeps merging until the ball is one connected blotch covering half the surface. So "about five spots" was a passing moment around step 200, not a property of the settings — and calibrating to it by choosing when to stop would have produced a number that looked right and meant nothing. The chemistry this campaign has always used makes *mazes*, not spots. Nobody noticed because nothing counted spots, and the movie's colour scale was rescaled every frame, so a merging pattern and a stable one looked identical in both the numbers and the pictures.
The calibration became a search for where stable spots actually live, tested by counting at two well-separated times. Twenty settings; the winner gives three spots that stay three spots across three independent runs, and is frozen as the default.

Phase 4b — Numerical stability: can a spec be integrated at all? IN PROGRESS

Why this became a phase, mid-flight. The pre-launch stopped on a control whose chemistry went non-finite. Chasing it found a stability bound that had been **50 times too permissive** for as long as the campaign has existed — and every ceiling derived from it inherited the same factor, so the search box advertised a region the integrator cannot enter.
The arithmetic. An explicit diffusion step is stable while $\mathrm{dt} \cdot \chi \cdot \max(d_a, d_h)$ stays below one. `translate` scales χ by $1/\mathrm{dt}$ — that is the clock fix, which exists because the chemistry was being integrated on the mechanics substep — so the engine steps $\mathrm{dt} \cdot (\chi/\mathrm{dt}) \cdot d$, and the dt cancels. The check multiplied by it anyway, against the *unscaled* χ, and so reported **0.056** for a composition whose real number is **2.8**.
Why nobody saw it. The bound was a *comment*: "CFL stays satisfied: $0.02 \times 65 \times 0.16 = 0.21$". True when written. Then the defaults moved and Phase 2 widened the diffusivity box to reach Okuda's $\varphi = 10$, and the arithmetic in the comment quietly stopped describing the code. A comment cannot be re-derived — only re-read, and nobody had cause to re-read it. *A limit that one phase widens and another relies on has to be a check.*
What it cost. `okuda_route` — the recipe named for the target — sat at 2.8 and passed every gate. Its chemistry diverged within a hundred frames, and the Biologist refused every run that used it. That refusal is how this was found at all: the premise gate caught a defect no metric would have shown.
Done

- The bound is computed, not commented: `diffusion_cfl` and both ceilings drop the cancelled dt, and `translate` refuses a spec whose *emitted* values breach it.
- Defaults moved to `coral_fixed_ball`'s point — chosen because it is *measured* to run, not

because it sits inside a box.

- `cfl_audit.py`: every spec in the repository, on both paths that produce one. 52 of 54 hand-written configs are stable; the two that are not are the two that break, and the one which actually ran went non-finite exactly as predicted.

Then it was certified against the engine, and the corrected bound is still wrong. Twenty specs on the cluster, 300 frames each, sweeping across the predicted wall. An argument about where a limit lies is worth nothing until the boundary it predicts is the boundary the engine has — and it is not:

CFL	χ	d_h	measured	
0.10	0.01	**10**	finite, patterns	**Okuda's** $\varphi = 10$ **IS reachable**
0.16	0.08	2	finite, patterns	
0.21–0.55	1.3	0.16–0.42	finite, patterns	the working band
0.48	3	0.16	finite, patterns	
0.50	0.05	10	non-finite	predicted stable
0.64	4	0.16	finite but INERT	the activator dies; P4
0.65	1.3	0.5	non-finite	
0.96	6	0.16	finite but INERT	every cell activated; P4

Two things follow, and both were wrong in what I told Cedric an hour earlier.

Okuda's $\varphi = 10$ is reachable after all. At $\chi = 0.01$ it runs finite and patterns. I had inferred from a short probe that it was not, and said so. The probe was 80 frames and tested $\chi = 0.05$, which does fail — the wall for $d_h = 10$ sits between 0.01 and 0.05, and I had generalised from the wrong side of it.

One number cannot express this wall. At $\chi = 1.3$ the chemistry survives to CFL 0.55 and dies at 0.65; at $\chi = 0.05$ it dies at 0.50. And there is a *second* failure mode the CFL says nothing about: at $\chi = 4$ and $\chi = 6$ the run stays perfectly finite and the activator goes to zero. A stability bound does not see that, the Biologist does — P4, “the chemistry was silently extinguished”. **Finite is not the same as alive**, and a bound that only forbids divergence admits a whole region of dead runs.

The bound is now two clauses, and both are measured. CFL ≤ 0.45 *and* $\chi \leq 3$. The second could not be derived — it came from watching fourteen points on the cluster — and it exists because there are *two* failure modes where the formula only knew about one. Divergence is the one a CFL forbids. Extinction is the other: at $\chi = 4$ and $\chi = 6$ the run stays perfectly finite and the activator goes to zero. χ multiplies both diffusivities, so raising it flattens the activator's own gradient as fast as the inhibitor's, and past about 3 the local peak cannot outrun its own spreading. A stable run that measures nothing is not caught by a stability bound; only the Biologist sees it, and only afterwards.

The limit sits below the first observed failure, which is conservative on purpose. The measurements do not admit a clean threshold: 0.48 and 0.55 were stable, 0.50 was not — and the 0.50 case is $d_h = 10$, so something beyond the product matters, since a large d_h fails where a small one survives at the same CFL. Until that term is derived the bound refuses two points known to be fine. Refusing a good composition costs a proposal; admitting a divergent one has cost a batch and a conclusion, twice.

Checked against all eight measured points: every bad one refused, no bad one admitted.

Still to do

- **Derive the missing term.** A single product cannot order the failures, and the bound is conservative until something explains why $d_h = 10$ behaves differently.

- **The reaction still has no bound**, deliberately — the candidate was measured and refuted — so a composition can pass the diffusion check and die through the reaction.
- **Longer runs may move the wall.** These were judged at 300 frames; a run that survives 300 and dies at 900 is still a run the gate let through.
- **One preset sits on the wrong side.** `round_44_base` ran healthily as a hand config at CFL 0.056 — 901 frames, 2983 cells, activator 0.993 — and compiles to 2.8 as a composition, because the clock fix scales χ by 50. Both are true, and they are not the same experiment: the archived run's chemistry had a fiftieth of the reaction time the corrected clock gives, which is precisely the defect the fix was written for. Keeping it as a trusted reference keeps a pre-fix run in the trusted set. `validate_space` is 54/56 on exactly this, and the decision is Cedric's.

Whose job this is. The **Critic**. It is deterministic, it runs before any compute, and it already refuses a composition that is illegal, unrunnable or already tried — "cannot be integrated" is the same kind of statement. It is not the Metrologist's, who certifies that an *instrument* measures what it claims, and not the Biologist's, who asks whether the *specimen* is a tissue. This one is about whether the arithmetic can be performed, which is a property of the proposal, and the proposal is what the Critic reads.

Phase 4c — Multi-spot initialisation: several tubes, not one IN PROGRESS

Why. Cedric: *"did you forget to run multiple tubes from now on, not a single one"*. He is right, and the note already said so — one protrusion per run is a single sample, and you cannot tell "this mechanism makes tubes" from "this run made a tube". Several spots give within-run replication for free, and it is what Okuda shows. Every control I built during the pre-launch used one spot or none.

Measured, six runs at the stable point, 300 frames:

seed fraction	act_max	activated fraction	
0.02	0.423	0.40	
0.06	0.501	0.37	three seeds, **identical to 3 d.p.**
0.15	0.598	0.12	
0.30	0.000	0.10	the pattern is extinguished (P4)

Two findings, one of them a problem. Seeding more widely does not give more pattern — it gives *less*, and past 0.15 it gives none at all: a seed covering a third of the shell leaves no room for the inhibitor to separate domains, so the whole field decays together. The useful band is narrow and sits below 0.15.

And the three seeds were bit-identical — 0.501, 0.501, 0.501. `general.seed` was written into every spec and read by *nothing*: both stochastic operators take a `seed` (the mesh jitter, and which cells are nucleated) and the translator passed neither. "Three independent runs" were one run repeated, so every spread this campaign has quoted across seeds described the floating-point reproducibility of a single trajectory.

Fixed, and the fix follows the working specs rather than my first idea. I offset the two operators' seeds, reasoning that jitter and nucleation are independent draws. True, and beside the point: `archive_rounded.py`, `archive_vh_rd_coral.py` and `run_tyssue_fig5.py` all pass *one shared* SEED to both, and a composition that reproduces an archived run has to draw the numbers that run drew. Departing from the convention would have made every reproduction check compare two different experiments. A spec that declares a seed no operator consumes is now refused outright.

Replication, measured for the first time — three seeds, 300 frames, same composition:

	act_max	**activated fraction**	**elongation**
before	0.501, 0.501, 0.501	0.374 ×3	1.006 ×3
after	0.501, 0.447, 0.483	0.407, 0.379, 0.374	1.006, 1.006, 1.005
	sd **0.022**	sd **0.015**	sd 0.0005

That last row is what a difference between two compositions now has to beat to mean anything. It did not exist yesterday, and every claim made without it was comparing single samples.

Phase 5 — The experiment ATTEMPTED — VOID

A 27-run batch ran on 31 July and produced no evidence at all. Four shape features, two directions each, three seeds, plus three nulls — the first batch in the campaign whose every instrument had been certified before it ran, and built to satisfy the ablation rule of Phase 3.

All 27 runs ended at exactly 1778 cells, flagged `saturated`, `valid_evidence: false`. That is the same ceiling that voided the overnight study the day before, in the same document, for the same reason. The launcher set no cell count and no buffer, so it inherited the default that produces 1778.

What this says, and it is not that the fix failed. The flag worked — every run is correctly marked as non-evidence, which is exactly what Phase 0 item 12 was for. What is missing is one step earlier: *nothing refused to start.* Detecting a spent batch afterwards and preventing it are different jobs, and only the first is built.

It also broke this phase's own stated rule — *no experiment until every instrument it will be read with exists and has been proved to work* — while Phase 0 still had two open items. Stating a rule and then not being stopped by it is the same class of problem as the buffer: a check that exists in prose but not in code.

Before it is re-run: a pre-launch gate that refuses any batch whose buffer cannot hold the cell count it is aiming at. It is arithmetic, it costs nothing, and it applies to faithful and exploratory runs alike.

What changes otherwise. Several seed spots rather than one, so a run carries its own replication. Baseline growth above zero — cells grow by taking up nutrients, and a protrusion drawn from existing material is an ablation, not a setting. Chemistry brought inside the search: the whole Turing half currently sits outside the loop's reach because the stage curriculum was written when the objective was "find a tube". And the starting conditions taken from the Grounder rather than from a default.

Runs locally. Two GPUs; a 260-frame run takes 52 seconds. The cluster is not needed and is currently off.

Phase 6 — The demonstration NOT STARTED

Goal. Run the designed grid — six to eight simulations — that produces the figure: tube, undulation and branching, at his settings.

What you get. *The figure.*

Phase 7 — The method claim NOT STARTED

Goal. Write up what the loop found by itself versus what we told it, with the record of predictions and retractions as the evidence.
What you get. The argument that Track A works, resting on Track B.

9. The 31 July lesson, in three lines

Ten defects in one day, every one found by Cedric looking at a picture and none by the loop — a ball that shrank to nothing, a movie where every cell turned green at once, a coral that began uniformly red. They were all the same kind: the loop could tell whether a *simulation* succeeded and had never been able to tell whether the *specimen* was a tissue. Phase 1 is the answer to exactly that, and the only way to find out whether it is enough is to run a batch and see whether the gate catches anything before Cedric does.

10. What I need from you

Nothing blocking. Two things worth knowing:

1. **The buffer gate is the one decision worth taking now.** Phase 5 has already been spent once on runs that could not have produced evidence. A pre-launch refusal costs nothing and would have saved the batch.
2. **Tell me if this note is the right shape.** Too long, too short, wrong order — it is cheap to change, and it is the thing that lets you steer.

Appendix — words I could not avoid

Only these four. Everything else above is ordinary English.

- **Cell sheet (or mesh)** — the model of the tissue: a hollow ball of polygonal cells sharing walls, like a football.
- **Operator** — one mechanism as a reusable building block: "cells divide", "chemical spreads", "surface pulls tight". A model is a combination of these.
- **The loop** — the automated cycle: propose a change, predict the outcome, run, measure, record whether the prediction held.
- **Phase** — one of the eight stages above. Our agreed stopping points.

The four measurements, renamed

Agreed 31 July. The old names misled in three separate ways and are being retired: `protr` reads as a length but is a *ratio*; `final` became ambiguous the moment we introduced the evidence horizon; and `Q` said nothing at all. The rename happens as one mechanical pass once the current work lands, with old records aliased on read — the archive is never rewritten.

was	is	means
`protr`	`elongation`	the 95th-percentile cell radius divided by the median. **1.0 is a sphere.** A ratio, not a length — reading 1.62 as an amount of protrusion is the mistake the old name invited.
`protr_peak`	`elongation_peak`	the most elongated it ever got, over *valid* frames only.
`protr_final`	`elongation_at_end`	at the last *valid* frame — not simply the last frame.
`Q`	`elongation_unforced`	what remains after growth and pushing are switched off and the tissue relaxes. **This is the forced-versus-grown test**, and it is the whole reason the campaign exists: we can already make a tube by pushing; the question is whether the tissue will hold one by itself.

My working log is `SESSION_LOG.md`; the findings register is `findings.py` and the premises are `PREMISES.md`. You should not need any of them.